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Studierichting: Informatica

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$$\begin{aligned} & \sum_{n=2}^{\infty} \frac{\ln n}{n^2} \\ &= \int_2^{+\infty} \frac{\ln x}{x^2} dx = - \int_2^{+\infty} \ln x d\left(\frac{1}{x}\right) \\ &= - \left[\frac{\ln x}{x} \right]_2^{+\infty} + \int_2^{+\infty} \frac{1}{x^2} dx \\ &= - \left[\frac{\ln x + 1}{x} \right]_2^{+\infty} \\ &= \lim_{a \rightarrow +\infty} -\frac{-\ln a - 1}{a} - \frac{\ln 2 - 1}{2} = 0 - \frac{-\ln 2 - 1}{2} = \frac{\ln 2 + 1}{2} \Rightarrow \text{bestaat} \\ & \begin{array}{l} \text{integraalcriterium} \\ \Rightarrow \end{array} \text{convergent} \end{aligned}$$

References